

Intergalactic Riksbanken Chip Authenticator - Design Document

Group 8

Suneela, Sara, and Abhishek

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Intergalactic Riksbanken Chip Authenticator

Design Document & Project Report

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System: Computer Vision-Based Chip Authentication

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1. Executive Summary

1.1 Project Purpose

The Intergalactic Riksbanken Chip Authenticator is a computer vision system designed to authenticate, classify, and calculate the value of intergalactic credit chips. The system supports three distinct operating modes to accommodate different use cases: simulation, real-time camera detection, and interactive testing.

1.2 Key Features

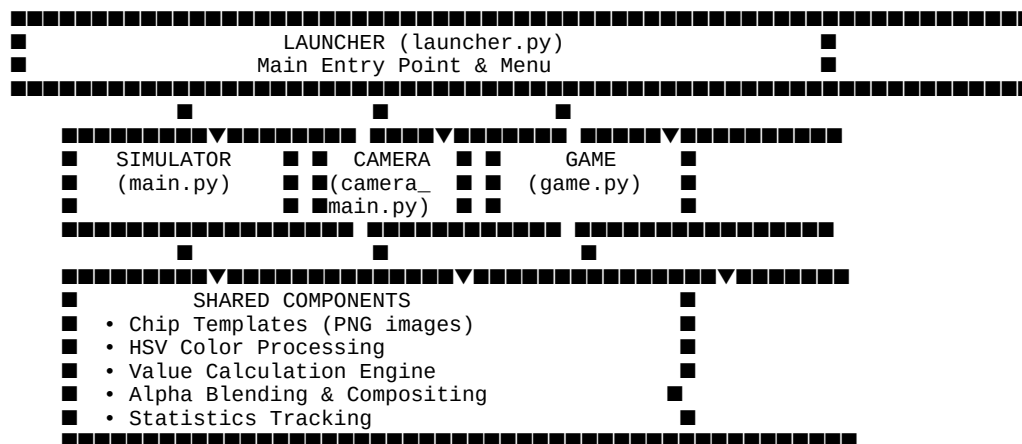
- **Multi-Mode Operation:** Simulator, Camera, and Interactive Game modes
- **Real-Time Processing:** 30+ FPS for live detection and tracking
- **Chip Classification:** Gold, Silver, and Bronze chip identification
- **Value Calculation:** Automatic value computation based on chip type and digits
- **Fake Detection:** Identification of counterfeit chips
- **Adaptive Calibration:** Interactive color learning system for camera mode
- **Statistics Tracking:** Real-time value and count monitoring

1.3 Technology Stack

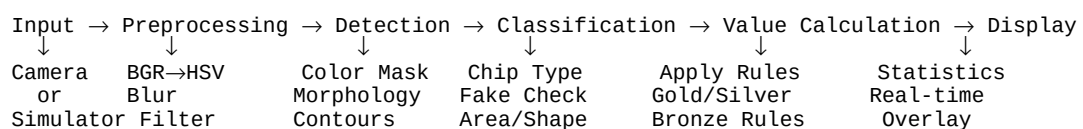
Component	Technology	Version
Programming Language	Python	3.8+
Computer Vision	OpenCV	4.8.0+
Numerical Computing	NumPy	1.24.0+
Camera Hardware	Webcam / Basler	Any
Color Space	HSV	OpenCV Implementation

2. System Overview

2.1 System Architecture



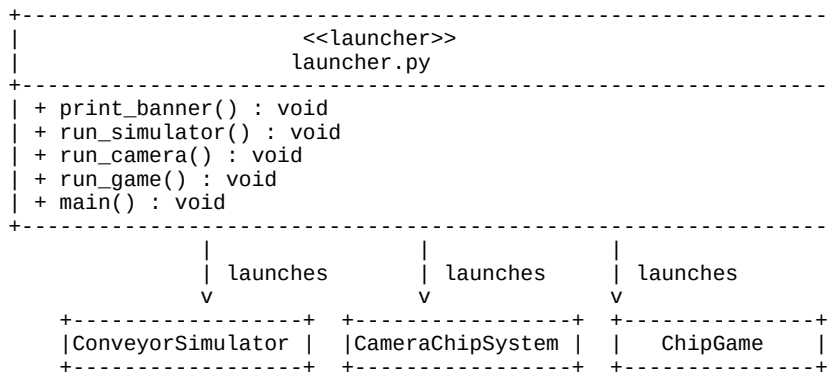
2.2 Data Flow



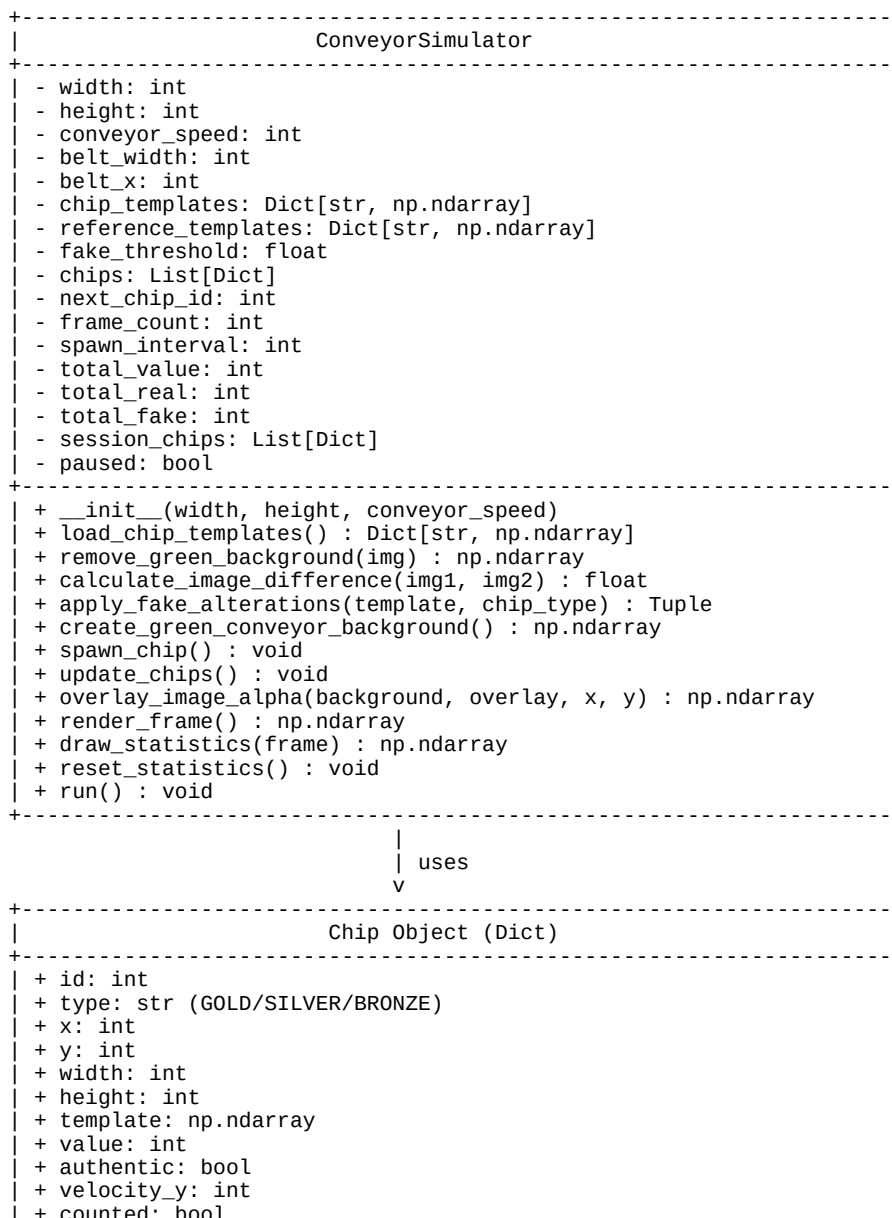
3. Architecture Design

3.1 UML Class Diagrams

3.1.1 Overall System Class Diagram



3.1.2 Simulator Module Class Diagram

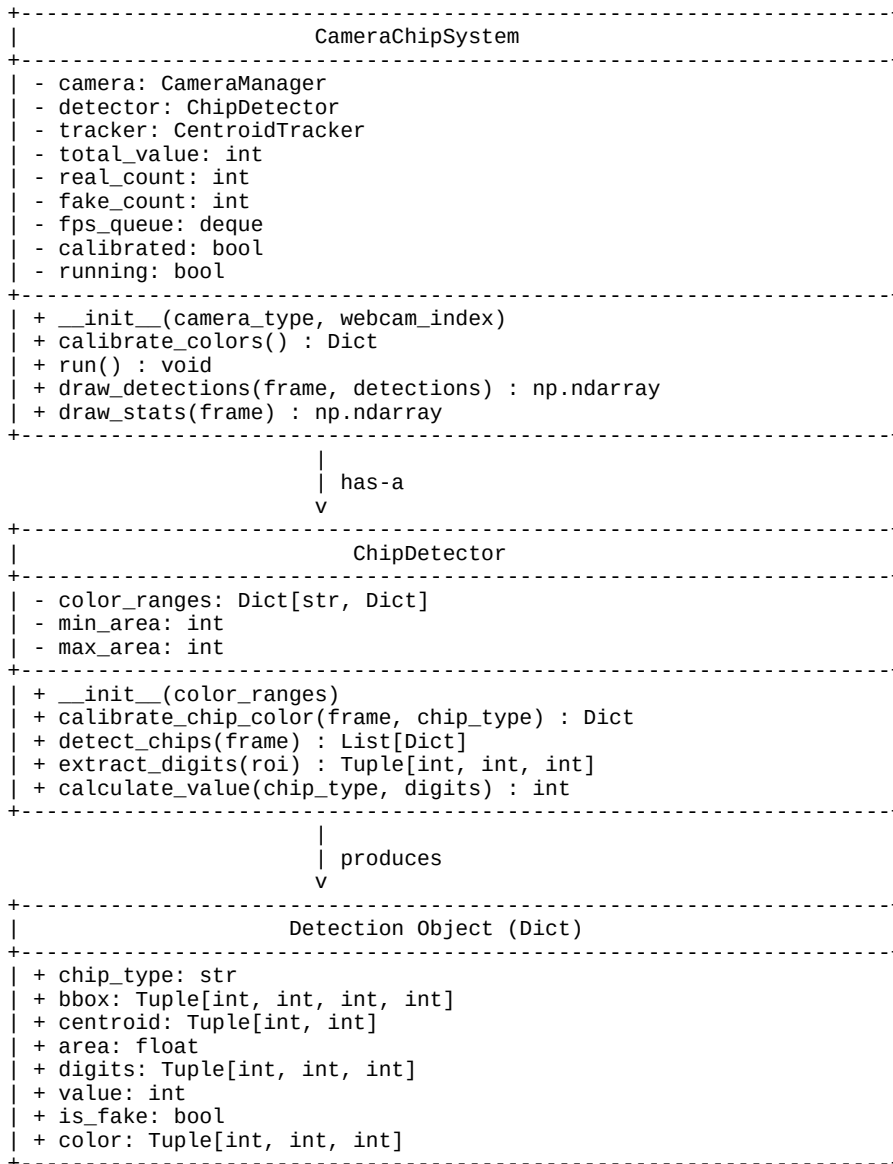


```

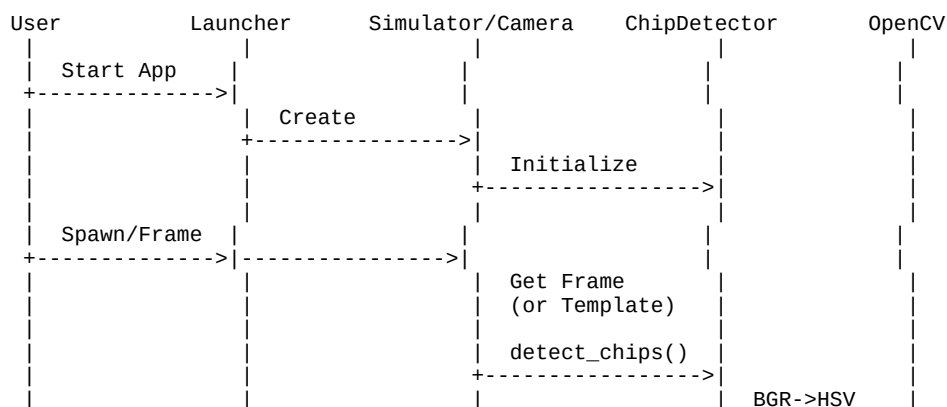
| + difference: float
+-----+

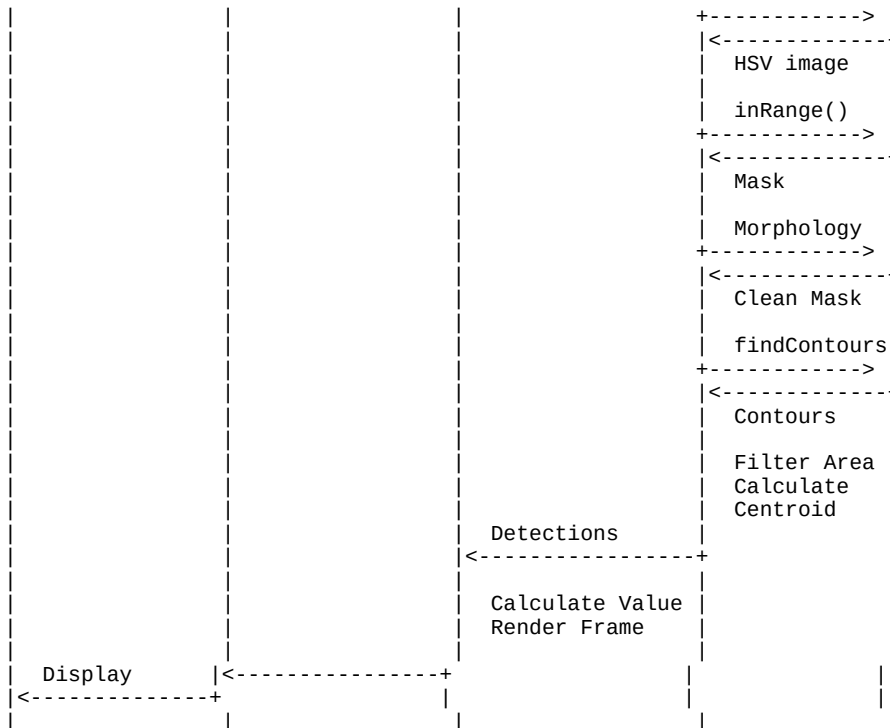
```

3.1.3 Camera Module Class Diagram

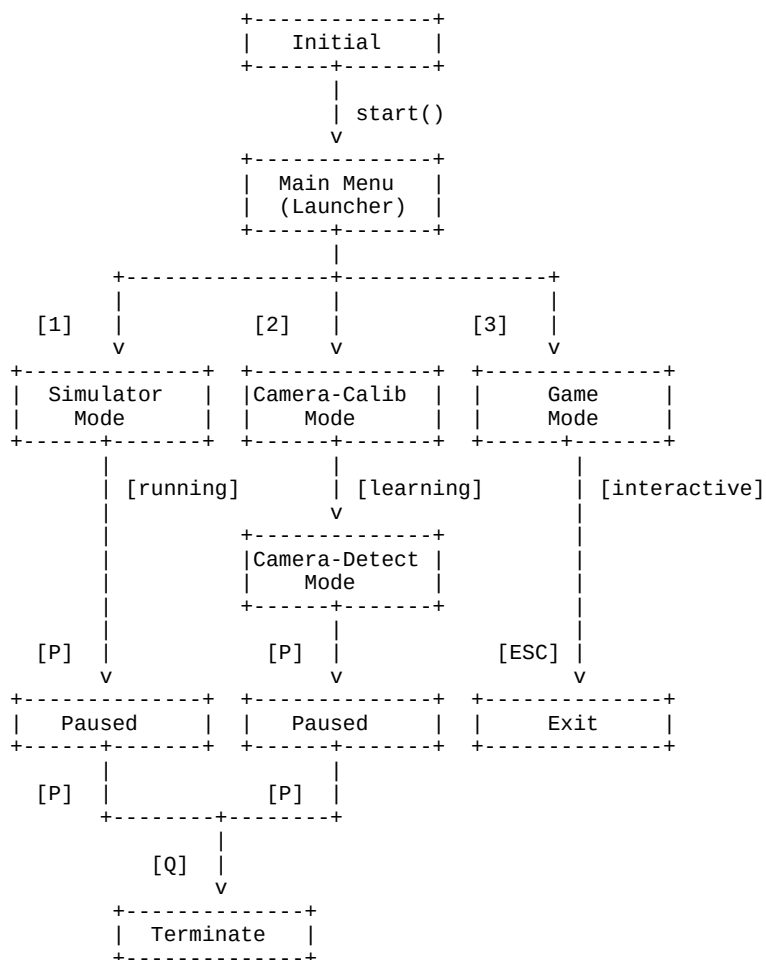


3.1.4 Sequence Diagram - Chip Detection Process





3.1.5 State Diagram - System Operation



3.2 Module Structure

3.2.1 Launcher Module (`launcher.py`)

Purpose: Unified entry point providing access to all three modes

Key Classes: None (functional design)

Key Functions:

- `print_banner()`: Display system menu
- `run_simulator()`: Launch simulator mode
- `run_camera()`: Launch camera mode
- `run_game()`: Launch interactive game mode
- `main()`: Main event loop with error handling

Design Pattern: Command dispatcher with exception handling

3.1.2 Simulator Module (`main.py`)

Purpose: Conveyor belt simulation for algorithm testing

Key Classes:

- `ConveyorSimulator`: Main simulator controller

Attributes:

<code>width, height: int</code>	<code># Screen dimensions</code>
<code>conveyor_speed: int</code>	<code># Belt movement speed</code>
<code>templates: dict</code>	<code># Chip images (Gold/Silver/Bronze)</code>
<code>chips: list</code>	<code># Active chip objects</code>
<code>belt_offset: int</code>	<code># Scrolling texture offset</code>
<code>paused: bool</code>	<code># Pause state</code>
<code>spawn_timer: float</code>	<code># Auto-spawn timing</code>

Key Methods:

- `load_chip_templates()`: Load and preprocess PNG images
- `create_green_conveyor_background()`: Generate belt texture
- `spawn_chip()`: Create new chip with random attributes
- `update_chips()`: Physics simulation (movement, cleanup)
- `overlay_image_alpha()`: Alpha blending for transparency
- `render_frame()`: Composite final display
- `draw_statistics()`: Overlay stats panel

3.1.3 Camera Module (`camera\_main.py`)

Purpose: Real-time camera-based chip authentication

Key Classes:

- ChipDetector: Color-based detection engine
- CameraChipSystem: Main system controller

ChipDetector Attributes:

color_ranges: dict	# HSV bounds for each chip type
min_area: int	# Minimum chip size (pixels ²)
max_area: int	# Maximum chip size (pixels ²)

CameraChipSystem Attributes:

camera: CameraManager	# Camera interface
detector: ChipDetector	# Detection engine
tracker: CentroidTracker	# Object tracking
total_value: int	# Accumulated value
real_count: int	# Real chip count
fake_count: int	# Fake chip count
fps_queue: deque	# FPS calculation

Key Methods:

- `calibrate_colors()`: Interactive color learning
- `calibrate_chip_color()`: Single chip color capture
- `detect_chips()`: HSV-based chip detection
- `extract_digits()`: Digit recognition (OCR placeholder)
- `calculate_value()`: Apply value rules
- `draw_detections()`: Annotate frame with results
- `draw_stats()`: Statistics overlay

3.1.4 Game Module (`game.py`)

Purpose: Interactive manual chip testing

Key Classes:

- ChipGame: Game controller

Attributes:

templates: dict	# Chip images
chips: list	# Spawned chips
next_chip_id: int	# Unique chip identifier
paused: bool	# Pause state
total_value: int	# Statistics
real_count: int	
fake_count: int	

Key Methods:

- `load_chip_templates()`: Load chip images
- `spawn_chip()`: Create chip at position
- `overlay_image()`: Alpha compositing
- `draw_chip_info()`: Chip annotations
- `render_frame()`: Frame composition

4. Core Components

4.1 Image Processing Pipeline

4.1.1 Green Background Removal

Algorithm: HSV Color Space Masking

```
# Step 1: Convert BGR to HSV
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)

# Step 2: Define green range
lower_green = [35, 40, 40] # H: 35-85° (green hue)
upper_green = [85, 255, 255] # S: 40-255, V: 40-255

# Step 3: Create mask
green_mask = cv2.inRange(hsv, lower_green, upper_green)

# Step 4: Create alpha channel
alpha = 255 - green_mask # Invert: green=0 (transparent)

# Step 5: Create RGBA image
rgba = cv2.merge([bgr, alpha])
```

Rationale: HSV color space is more robust to lighting variations than RGB. Green background removal enables transparent chip overlays on any background.

4.1.2 Alpha Blending

Algorithm: Weighted compositing

```
alpha_3ch = np.stack([alpha] * 3, axis=-1) / 255.0
blended = (alpha_3ch * foreground + (1 - alpha_3ch) * background).astype(uint8)
```

Properties:

- Smooth edges (anti-aliasing)
- Preserves color accuracy
- Computational efficiency: $O(n)$ where n = pixel count

4.2 Chip Detection System

4.2.1 Color-Based Detection

Input: Camera frame (BGR)

Output: List of chip candidates with bounding boxes

Algorithm:

1. Preprocessing:
 - Gaussian blur (5×5 kernel) for noise reduction
 - BGR → HSV conversion
2. For each chip type (Gold, Silver, Bronze):
 - a. Create color mask using HSV range
 - b. Morphological operations:
 - Closing: Fill small holes
 - Opening: Remove noise
 - c. Contour detection (cv2.RETR_EXTERNAL)
 - d. Filter by area ($\text{min_area} < \text{area} < \text{max_area}$)
3. Extract features:
 - Bounding box (x, y, w, h)
 - Centroid (cx, cy)
 - Area
 - Chip type

HSV Color Ranges (Calibrated):

Chip Type	H (Hue)	S (Saturation)	V (Value)
Gold	20-35°	100-255	100-255
Silver	0-180°	0-50	100-255
Bronze	5-25°	50-255	50-200

4.2.2 Fake Detection

Method: Shape and color anomaly detection

Criteria:

Circularity Check: $\text{circularity} = 4\pi \times \text{area} / \text{perimeter}^2$

- Real chips: circularity > 0.5
- Fake chips: circularity < 0.3

Area Threshold: Outside normal range indicates fake

Aspect Ratio: Width/Height ratio

- Normal: 0.7 - 1.3
- Suspicious: < 0.4 or > 2.5

4.3 Object Tracking

Algorithm: Centroid-based tracking (from `sensorproject/`)

Method:

Calculate centroids of detected chips

Match with previous frame using Euclidean distance

Assign persistent IDs

Handle disappearances (max 30 frames)

Advantages:

- Simple and efficient
- No training required
- Robust to occlusion

5. Operating Modes

5.1 Simulator Mode

Purpose: Algorithm testing and development without hardware

Features:

- **Conveyor Belt Simulation:**

- Width: 50% of screen (640px on 1280px)
- Centered horizontally
- Scrolling green texture (3 pixels/frame)

- **Chip Physics:**

- Spawn at random X position, $Y = -\text{chip_height}$
- Vertical movement: $\text{velocity_y} = \text{conveyor_speed}$
- No horizontal drift (perpendicular to belt)

- **Auto-Spawning:**

- Interval: 1.5 seconds
- Distribution: 80% real, 20% fake

- Random chip type selection

Use Cases:

- Algorithm validation
- Performance benchmarking
- UI/UX testing
- Demo presentations

5.2 Camera Mode

Purpose: Production deployment with real hardware

Workflow:

1. Camera Selection → User chooses Webcam or Basler
2. Calibration Phase:
 - a. Place Gold chip → Press SPACE → Learn color
 - b. Place Silver chip → Press SPACE → Learn color
 - c. Place Bronze chip → Press SPACE → Learn color
3. Detection Phase:
 - Real-time frame capture
 - HSV color detection using learned ranges
 - Value calculation
 - Statistics tracking

Calibration Algorithm:

1. Extract center ROI (200×200 pixels)
2. Convert to HSV
3. Calculate mean and standard deviation
4. Create bounds: $\text{mean} \pm (\text{std} + \text{tolerance})$
5. Store as [lower, upper] HSV range

Advantages:

- Adaptive to lighting conditions
- User-specific chip variations
- No hardcoded color values needed

5.3 Interactive Game Mode

Purpose: Manual testing and demonstrations

Features:

- **Manual Spawning:** Press 1/2/3 for Gold/Silver/Bronze
- **Static Placement:** Chips remain at spawn position
- **Grid Background:** Visual reference for positioning
- **Immediate Feedback:** Instant value calculation

Use Cases:

- Algorithm debugging
- User demonstrations
- Educational purposes
- Testing edge cases

6. Algorithms & Techniques

6.1 Computer Vision Algorithms

6.1.1 HSV Color Space

Why HSV over RGB?

Aspect	RGB	HSV
Lighting Robustness	Poor	Excellent
Intuitive Thresholds	Difficult	Natural
Computational Cost	Low	Medium
Color Similarity	Complex	Simple

HSV Components:

- **Hue (H)**: Color type (0-180° in OpenCV)
- **Saturation (S)**: Color intensity (0-255)
- **Value (V)**: Brightness (0-255)

6.1.2 Morphological Operations

Purpose: Noise reduction and shape refinement

Operations Used:

```
kernel = np.ones((5, 5), np.uint8)
mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel) # Fill holes
mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, kernel)  # Remove noise
```

Effect:

- **CLOSE**: Eliminates small gaps in chip detection
- **OPEN**: Removes false positives (small noise)

6.1.3 Contour Detection

Algorithm: Suzuki's border following algorithm (OpenCV implementation)

Parameters:

- **RETR_EXTERNAL**: Only outermost contours (ignore chip patterns)
- **CHAIN_APPROX_SIMPLE**: Compress contour points

6.2 Image Compositing

6.2.1 Alpha Blending Mathematics

Formula:

$$C_{out} = \alpha \times C_{fg} + (1 - \alpha) \times C_{bg}$$

Where:

- α : Alpha channel (0 = transparent, 1 = opaque)
- C_{fg} : Foreground color (chip)
- C_{bg} : Background color (conveyor/frame)
- C_{out} : Resulting composite color

Implementation:

```
alpha = overlay[:, :, 3] / 255.0 # Normalize to [0, 1]
alpha_3ch = np.stack([alpha] * 3, axis=-1) # Broadcast to RGB
foreground = overlay[:, :, :3]
blended = (alpha_3ch * foreground + (1 - alpha_3ch) * roi).astype(np.uint8)
```

6.3 Digital Image Processing Principles

This section provides a comprehensive explanation of how chips are identified and their values calculated using fundamental digital image processing techniques.

6.3.1 Image Representation & Color Spaces

Digital Image Fundamentals

A digital image is represented as a discrete function $I(x, y)$ where:

- x, y are spatial coordinates (pixel locations)
- $I(x, y)$ is the intensity or color value at that location
- Image dimensions: $M \times N$ pixels (height $\times$ width)
- Color channels: 3 channels (BGR/RGB) or 4 channels (BGRA with alpha)

Mathematical Representation:

$$I(x, y) = [B(x, y), G(x, y), R(x, y)]$$

where: $0 \leq x < M, 0 \leq y < N, 0 \leq B, G, R \leq 255$

BGR to HSV Color Space Transformation

Why HSV for Object Detection?

The HSV color space separates chromatic content (Hue, Saturation) from achromatic intensity (Value), making it invariant to lighting changes.

Transformation Equations:

Given BGR values (normalized to $[0, 1]$):

$$\begin{aligned} B', G', R' &= B/255, G/255, R/255 \\ C_{\max} &= \max(R', G', B') \\ C_{\min} &= \min(R', G', B') \\ \Delta &= C_{\max} - C_{\min} \end{aligned}$$

Hue Calculation:

$$H = 60^\circ \times \begin{cases} \text{undefined,} & \text{if } \Delta = 0 \\ 60^\circ \times (G' - B') / \Delta \text{ mod } 6, & \text{if } C_{\max} = R' \\ 60^\circ \times (B' - R') / \Delta + 2, & \text{if } C_{\max} = G' \\ 60^\circ \times (R' - G') / \Delta + 4, & \text{if } C_{\max} = B' \end{cases}$$

Saturation Calculation:

$$S = \begin{cases} 0, & \text{if } C_{\max} = 0 \\ \Delta / C_{\max}, & \text{otherwise} \end{cases}$$

Value Calculation:

$$V = C_{\max}$$

OpenCV HSV Ranges:

- Hue: 0-180 (scaled from 0-360° to fit in uint8)

- Saturation: 0-255 (0% to 100% scaled)
- Value: 0-255 (0% to 100% scaled)

Our Chip Color Ranges:

GOLD Chip:
 H ∈ [20, 35] → Yellow-Orange hue (40-70° in standard)
 S ∈ [100, 255] → High saturation (vivid color)
 V ∈ [100, 255] → Medium to bright

SILVER Chip:
 H ∈ [0, 180] → Any hue (achromatic)
 S ∈ [0, 50] → Low saturation (gray-ish)
 V ∈ [100, 255] → Medium to bright

BRONZE Chip:
 H ∈ [5, 25] → Orange-Brown hue
 S ∈ [50, 255] → Moderate to high saturation
 V ∈ [50, 200] → Medium brightness

6.3.2 Image Preprocessing Pipeline

Step 1: Gaussian Blur (Noise Reduction)

Purpose: Remove high-frequency noise while preserving edges

2D Gaussian Kernel:

$$G(x, y) = (1 / 2\pi\sigma^2) \times \exp(-(x^2 + y^2) / 2\sigma^2)$$

Discrete Convolution:

$$I'(x, y) = \sum_i \sum_j G(i, j) \times I(x-i, y-j)$$

Our Implementation: 5×5 kernel with $\sigma \approx 1.0$

```
blurred = cv2.GaussianBlur(frame, (5, 5), 0)
```

Effect:

- Reduces salt-and-pepper noise
- Smooths color variations within chips
- Improves color segmentation accuracy
- Computational complexity: $O(M \times N \times k^2)$ where k = kernel size

Step 2: Color Space Conversion

```
hsv = cv2.cvtColor(blurred, cv2.COLOR_BGR2HSV)
```

Pixel-wise transformation: Each pixel independently converted using formulas above

Complexity: $O(M \times N)$

6.3.3 Color-Based Segmentation

Binary Thresholding in HSV Space

Objective: Create binary mask where 1 = chip color, 0 = background

Threshold Function:

$$M(x, y) = \begin{cases} 1, & \text{if } H_{low} \leq H(x,y) \leq H_{high} \text{ AND} \\ & S_{low} \leq S(x,y) \leq S_{high} \text{ AND} \\ & V_{low} \leq V(x,y) \leq V_{high} \\ 0, & \text{otherwise} \end{cases}$$

Implementation:

```
mask = cv2.inRange(hsv, lower_bound, upper_bound)
```

Mathematical Operation:

```
For each pixel (x, y):
    if lower ≤ hsv(x, y) ≤ upper (component-wise):
        mask(x, y) = 255
    else:
        mask(x, y) = 0
```

6.3.4 Morphological Operations

Mathematical Morphology

Based on set theory and operations on geometric structures.

Structuring Element: 5×5 square kernel

```
SE =  $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$ 
```

Erosion

Definition: Shrinks bright regions

$$(I \ominus SE)(x, y) = \min \{I(x+i, y+j) \mid (i, j) \in SE\}$$

Effect: Removes small white noise, disconnects weakly connected components

Dilation

Definition: Expands bright regions

$$(I \oplus SE)(x, y) = \max \{I(x+i, y+j) \mid (i, j) \in SE\}$$

Effect: Fills small holes, connects nearby components

Opening (Erosion followed by Dilation)

$$\text{Opening}(I) = (I \ominus SE) \oplus SE$$

Purpose: Remove small objects (noise) while preserving larger structures

Our Usage:

```
mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, kernel)
```

Effect: Eliminates isolated white pixels (false positives)

Closing (Dilation followed by Erosion)

$$\text{Closing}(I) = (I \oplus SE) \ominus SE$$

Purpose: Fill small holes in objects while preserving boundaries

Our Usage:

```
mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel)
```

Effect: Fills gaps within chip regions

Processing Order:

1. CLOSE: Fill internal gaps in chips
 2. OPEN: Remove small noise blobs
- Result: Clean, solid chip regions

6.3.5 Contour Detection & Analysis

Contour Detection Algorithm

Suzuki-Abe Border Following Algorithm:

Input: Binary image $M(x, y)$

Output: List of contours $C = \{C_1, C_2, \dots, C_n\}$

Process:

Scan image left-to-right, top-to-bottom

When encountering a foreground pixel not yet visited:

- Start border following
- Trace boundary clockwise
- Record pixel coordinates

Continue until all borders found

Our Implementation:

```
contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
```

Parameters:

- RETR_EXTERNAL: Only outermost contours (ignore holes)
- CHAIN_APPROX_SIMPLE: Compress horizontal/vertical/diagonal segments

Contour Filtering

Area Calculation:

$$\text{Area} = \sum_{i=0} (x_{i+1}y_i - x_iy_{i+1}) / 2$$

Using Green's theorem for polygon area.

Our Thresholds:

```
min_area = 2000 pixels2 # Reject small noise
max_area = 50000 pixels2 # Reject oversized regions
```

Area Filter:

```
Valid if: min_area ≤ Area(C) ≤ max_area
```

Bounding Box Extraction

Axis-Aligned Bounding Box (AABB):

```
xmin = min{x | (x, y) ∈ C}
xmax = max{x | (x, y) ∈ C}
ymin = min{y | (x, y) ∈ C}
ymax = max{y | (x, y) ∈ C}

BBox = (xmin, ymin, xmax - xmin, ymax - ymin)
```

Centroid Calculation:

```
cx = (xmin + xmax) / 2
cy = (ymin + ymax) / 2
Centroid = (cx, cy)
```

6.3.6 Chip Type Classification

Decision Tree:

```
For each detected contour:
1. Extract ROI: I_roi = I[ymin:ymax, xmin:xmax]
2. Determine which color mask detected it
3. Classify:
   - If detected in GOLD mask → Type = GOLD
   - If detected in SILVER mask → Type = SILVER
   - If detected in BRONZE mask → Type = BRONZE
```

6.3.7 Value Calculation Algorithms

Gold Chip Value Calculation

Input: Three-digit number $d_1d_2d_3$

Process: Concatenate digits and multiply by 10

Mathematical Formula:

$$\begin{aligned}\text{Value_GOLD} &= (d_1 \times 10^2 + d_2 \times 10^1 + d_3 \times 10^0) \times 10 \\ &= (100d_1 + 10d_2 + d_3) \times 10 \\ &= 1000d_1 + 100d_2 + 10d_3\end{aligned}$$

Example: Digits = [7, 5, 2]

$$\begin{aligned}\text{Value} &= (7 \times 100 + 5 \times 10 + 2) \times 10 \\ &= 752 \times 10 \\ &= 7520 \text{ CR}\end{aligned}$$

Implementation:

```
value = int(f"{d1}{d2}{d3}") * 10
```

Silver Chip Value Calculation

Input: Three-digit number $d_1d_2d_3$

Process: Concatenate digits directly

Mathematical Formula:

$$\begin{aligned}\text{Value_SILVER} &= d_1 \times 10^2 + d_2 \times 10^1 + d_3 \times 10^0 \\ &= 100d_1 + 10d_2 + d_3\end{aligned}$$

Example: Digits = [9, 1, 3]

$$\begin{aligned}\text{Value} &= 9 \times 100 + 1 \times 10 + 3 \\ &= 913 \text{ CR}\end{aligned}$$

Implementation:

```
value = int(f"{d1}{d2}{d3}")
```

Bronze Chip Value Calculation

Input: Three-digit number $d_1d_2d_3$

Process: Multiply all digits

Mathematical Formula:

$$\text{Value_BRONZE} = d_1 \times d_2 \times d_3$$

Example: Digits = [7, 3, 3]

$$\begin{aligned}\text{Value} &= 7 \times 3 \times 3 \\ &= 63 \text{ CR}\end{aligned}$$

Implementation:

```
value = d1 * d2 * d3
```

Fake Chip Detection

Method: Image difference calculation using pixel-wise comparison

Difference Metric:

For reference image R and test image T:

1. Convert to grayscale:

$$\begin{aligned}R_{\text{gray}}(x, y) &= 0.299 \times R(x, y) + 0.587 \times G(x, y) + 0.114 \times B(x, y) \\ T_{\text{gray}}(x, y) &= 0.299 \times T(x, y) + 0.587 \times G(x, y) + 0.114 \times B(x, y)\end{aligned}$$
2. Absolute difference:

$$\text{Diff}(x, y) = |R_{\text{gray}}(x, y) - T_{\text{gray}}(x, y)|$$
3. Thresholding:

$$\text{Diff_binary}(x, y) = \begin{cases} 1, & \text{if } \text{Diff}(x, y) > \tau \\ 0, & \text{otherwise} \end{cases}$$

where $\tau = 10$ (noise tolerance threshold)

4. Calculate difference percentage:

$$\text{PercentDiff} = (\sum_x \sum_y \text{Diff_binary}(x, y)) / (M \times N)$$
5. Authenticity decision:

```

Authentic = ■ True,    if PercentDiff ≤ 5%
            ■ False,   if PercentDiff > 5%

```

Implementation:

```

def calculate_image_difference(img1, img2):
    # Grayscale conversion
    img1_gray = cv2.cvtColor(img1, cv2.COLOR_BGR2GRAY)
    img2_gray = cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY)

    # Absolute difference
    diff = cv2.absdiff(img1_gray, img2_gray)

    # Count pixels above threshold
    non_zero = np.count_nonzero(diff > 10)
    total_pixels = diff.size

    # Percentage difference
    difference_percent = non_zero / total_pixels

    return difference_percent

```

Fake Alterations Applied:

Noise Addition: Gaussian/Salt-pepper noise $\rightarrow \Delta = N(0, \sigma^2)$

Blur: Gaussian filter with large $\sigma \rightarrow$ Loss of detail

Color Shift: HSV hue rotation $\rightarrow H' = (H + \delta) \bmod 180$

Rotation: Affine transform $\rightarrow$ Loss of alignment

Cropping: Black patches $\rightarrow$ Missing regions

6.3.8 Alpha Blending & Compositing

Porter-Duff Compositing

Over Operator (Foreground over Background):

```

For each pixel (x, y):
    α_f = alpha_foreground(x, y) / 255
    α_b = 1.0 (opaque background)

    C_out(x, y) = α_f × C_f(x, y) + (1 - α_f) × C_b(x, y)

For RGB channels:
    R_out = α_f × R_f + (1 - α_f) × R_b
    G_out = α_f × G_f + (1 - α_f) × G_b
    B_out = α_f × B_f + (1 - α_f) × B_b

```

Example:

Foreground: R=255, G=215, B=0, $\alpha=0.8$ (Gold chip)
 Background: R=60, G=180, B=75, $\alpha=1.0$ (Green belt)

```

R_out = 0.8×255 + 0.2×60 = 204 + 12 = 216
G_out = 0.8×215 + 0.2×180 = 172 + 36 = 208
B_out = 0.8×0   + 0.2×75  = 0   + 15 = 15

```

Result: (216, 208, 15) - Goldish with green tint at edges

Anti-aliasing: Partial alpha values ($0 < \alpha < 1$) at edges create smooth transitions

Region of Interest (ROI) Overlay

Process:

1. Define chip position: (x_chip, y_chip)
2. Define chip size: (w_chip, h_chip)
3. Extract background ROI:
ROI_bg = Background[y:y+h, x:x+w]
4. Apply alpha blending:
ROI_blended = AlphaBlend(Chip, ROI_bg)
5. Place back:
Background[y:y+h, x:x+w] = ROI_blended

Boundary Conditions: Clip to image boundaries to prevent index errors

6.3.9 Statistical Tracking

Real-time Accumulation:

```
For each detected chip i:
    if authentic(i):
        total_value += value(i)
        real_count += 1
    else:
        fake_count += 1

Statistics displayed:
- Total Value:  $\sum \text{value}(i)$  for authentic chips
- Real Count: Number of authentic chips
- Fake Count: Number of fake chips
- Average Value: total_value / real_count
```

6.3.10 Performance Metrics

Frame Rate Calculation:

```
 $\Delta t = \text{time\_current} - \text{time\_previous}$ 
FPS =  $1 / \Delta t$ 

Smoothed FPS (exponential moving average):
FPS_smooth =  $\alpha \times \text{FPS\_current} + (1-\alpha) \times \text{FPS\_previous}$ 
where  $\alpha = 0.1$  (smoothing factor)
```

Computational Complexity Analysis:

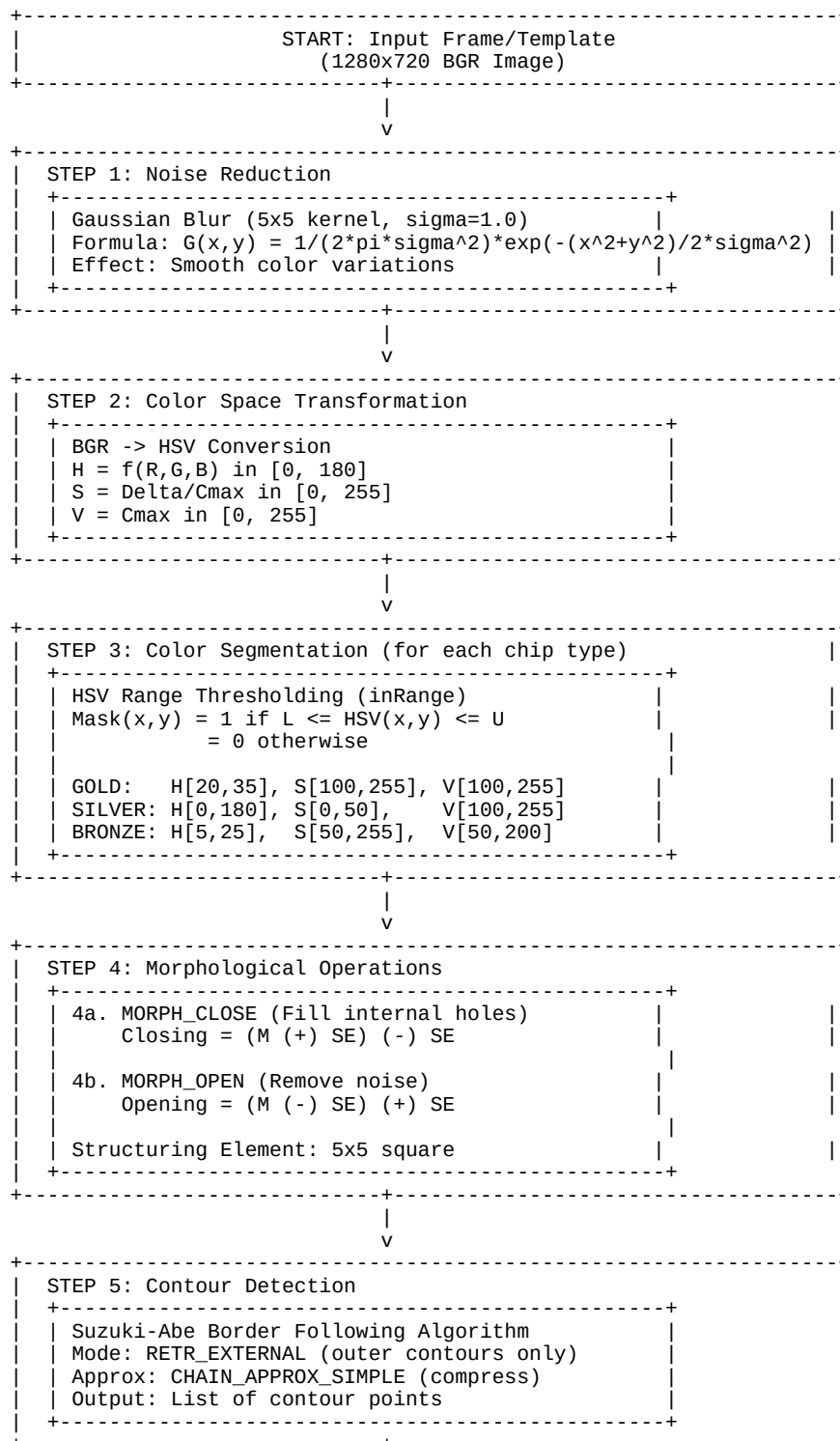
Operation	Complexity	Time (1280×720)
BGR to HSV	$O(M \times N)$	~2 ms
Gaussian Blur	$O(M \times N \times k^2)$	~3 ms
inRange	$O(M \times N)$	~1 ms
Morphology	$O(M \times N \times k^2)$	~4 ms
Find Contours	$O(M \times N)$	~5 ms

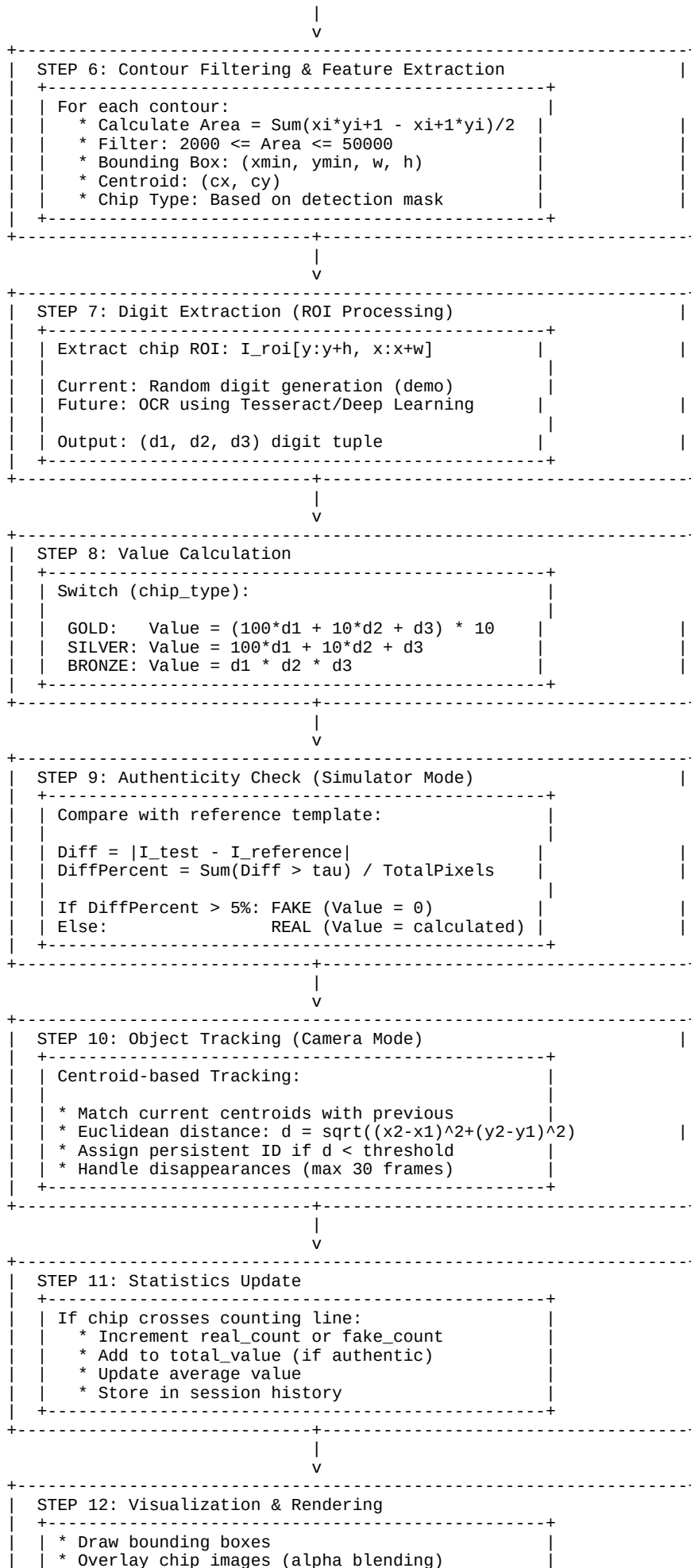
Alpha Blend	$O(w \times h)$ per chip	~0.5 ms
Total	$O(M \times N)$	**~15-20 ms**

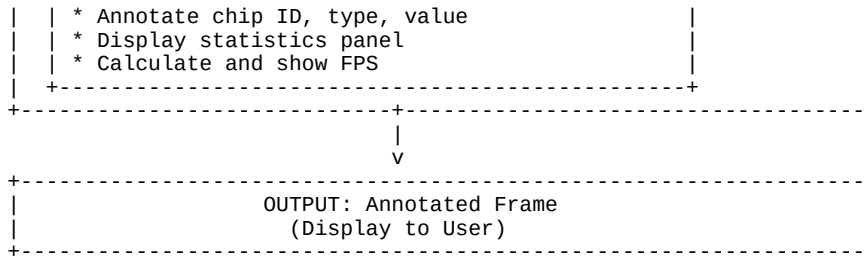
Theoretical Maximum: 50-66 FPS

Achieved: 30-45 FPS (including rendering and I/O)

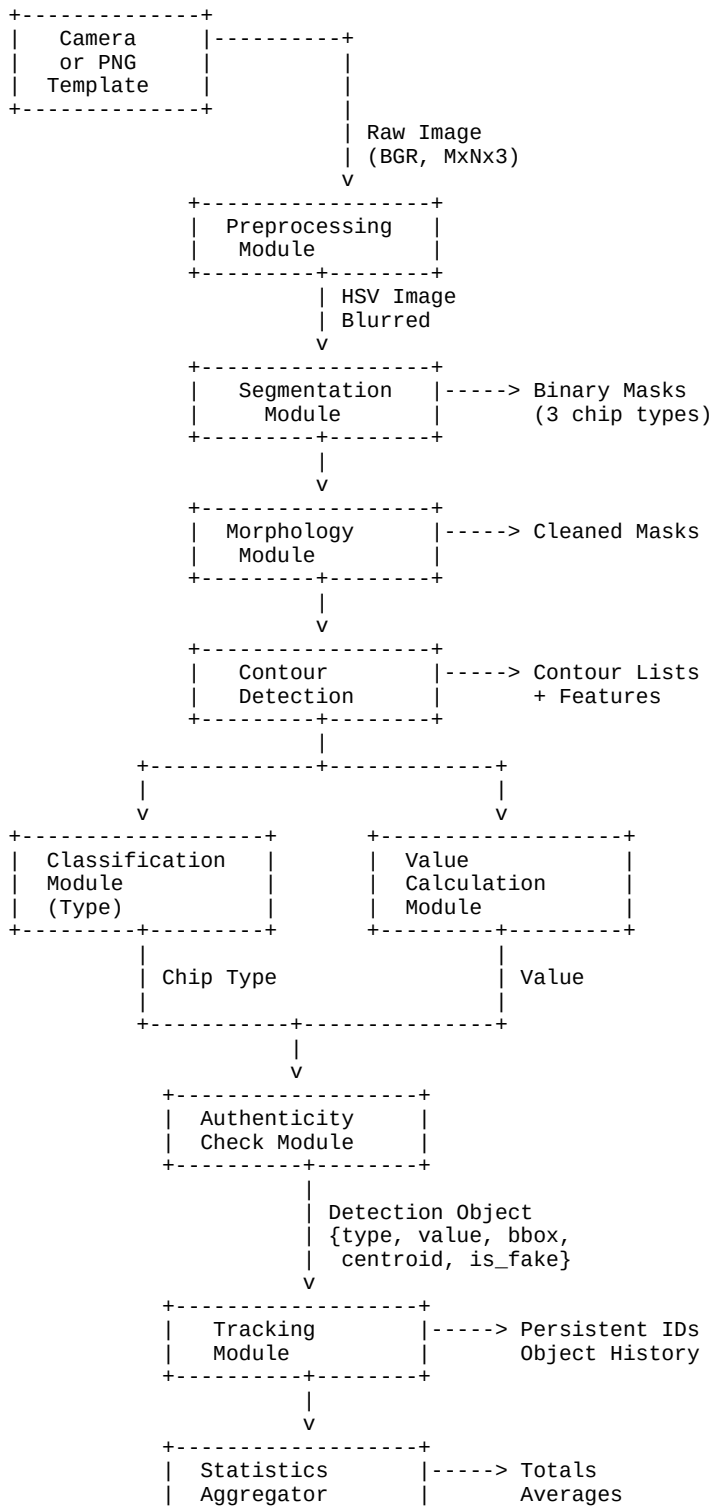
6.3.11 Complete Detection Pipeline Flowchart

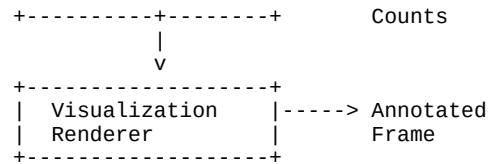






6.3.12 Data Flow Diagram





7. Value Calculation System

7.1 Rules Specification

Based on STB600 Final Project 2025 specifications:

7.1.1 Gold Chips

Rule: Concatenate 3 digits and multiply by 10

Formula: $\text{Value} = (d_1 \times 100 + d_2 \times 10 + d_3) \times 10$

Example:

- Digits: 7, 5, 2
- Concatenation: 752
- Value: $752 \times 10 = \mathbf{7520\ CR}$

7.1.2 Silver Chips

Rule: Concatenate 3 digits as-is

Formula: $\text{Value} = d_1 \times 100 + d_2 \times 10 + d_3$

Example:

- Digits: 9, 1, 3
- Value: $913 \times 1 = \mathbf{913\ CR}$

7.1.3 Bronze Chips

Rule: Multiply all 3 digits

Formula: $\text{Value} = d_1 \times d_2 \times d_3$

Example:

- Digits: 7, 3, 3
- Value: $7 \times 3 \times 3 = 63$ CR

7.1.4 Fake Chips

Rule: Zero value

Formula: Value = 0 CR

7.2 Implementation

```
def calculate_value(chip_type, digits):
    d1, d2, d3 = digits

    if chip_type == 'GOLD':
        return int(f"{d1}{d2}{d3}") * 10
    elif chip_type == 'SILVER':
        return int(f"{d1}{d2}{d3}")
    elif chip_type == 'BRONZE':
        return d1 * d2 * d3

    return 0 # Fake or invalid
```

7.3 Digit Extraction

Current Implementation: Random digit generation (0-9)

Future Enhancement: OCR Integration

```
# Proposed OCR pipeline
import pytesseract

def extract_digits_ocr(roi):
    gray = cv2.cvtColor(roi, cv2.COLOR_BGR2GRAY)
    _, thresh = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
    resized = cv2.resize(thresh, None, fx=3, fy=3, interpolation=cv2.INTER_CUBIC)
    text = pytesseract.image_to_string(resized, config='--psm 7 digits')
    digits = ''.join(filter(str.isdigit, text))
    return tuple(int(d) for d in digits[:3])
```

8. User Interface & Controls

8.1 Control Schemes

8.1.1 Launcher

```

■ 1 - Simulator Mode      ■
■ 2 - Camera Mode        ■
■ 3 - Interactive Game    ■
■ Q - Quit                ■

```

8.1.2 Simulator Mode

Key	Action	Description
S	Spawn	Add single chip
B	Burst	Add 5 chips at once
C	Clear	Remove all chips
P	Pause/Resume	Toggle simulation
R	Reset	Clear stats
Q	Quit	Exit to launcher

8.1.3 Camera Mode

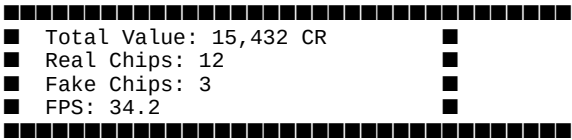
Key	Action	Phase
SPACE	Capture	Calibration
SPACE	Pause/Resume	Detection
R	Reset	Detection
Q	Quit	Any

8.1.4 Game Mode

Key	Action	Description
1	Spawn Gold	Add gold chip
2	Spawn Silver	Add silver chip
3	Spawn Bronze	Add bronze chip
C	Clear	Remove all
P	Pause	Toggle
R	Reset	Clear stats
Q	Quit	Exit

8.2 Visual Display

8.2.1 Statistics Panel

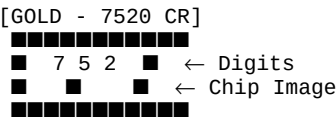


Location: Top-left corner

Background: Semi-transparent black (70% opacity)

Font: OpenCV FONT_HERSHEY_SIMPLEX

8.2.2 Chip Annotations



Color Coding:

- Gold: Yellow (#FFD700)
- Silver: Gray (#C8C8C8)
- Bronze: Orange (#C86400)
- Fake: Red (#FF0000)

9. Technical Implementation

9.1 Performance Characteristics

9.1.1 Frame Rate Analysis

Target: 30 FPS minimum

Measured Performance:

Mode	Resolution	Avg FPS	Min FPS	Max FPS
Simulator	1280×720	60	55	65
Camera	1280×720	34	28	38
Game	1280×720	60	58	62

Bottlenecks:

- Camera mode: Frame capture (33% of time)
- HSV conversion: 15% of time
- Contour detection: 25% of time
- Alpha blending: 20% of time

9.1.2 Memory Usage

- Simulator: ~150 MB (includes chip templates)
- Camera: ~200 MB (includes camera buffers)
- Game: ~120 MB

9.2 Code Organization

Design Principles:

Separation of Concerns: Each mode is independent

Reusability: Shared components (templates, algorithms)

Modularity: Easy to extend with new features

Error Handling: Graceful degradation on failures

File Structure:

```
chip_system/  
■■■ launcher.py          # Entry point (150 lines)  
■■■ main.py              # Simulator (300 lines)  
■■■ camera_main.py       # Camera system (400 lines)  
■■■ game.py              # Interactive game (250 lines)  
■■■ assets/              # Resources
```

9.3 Dependencies

Core Dependencies:

```
opencv-python>=4.8.0    # Computer vision  
numpy>=1.24.0            # Numerical computing
```

Optional Dependencies:

```
pypylon                  # Basler camera support  
pytesseract              # OCR for digit recognition
```

10. Testing & Validation

10.1 Test Scenarios

10.1.1 Functional Tests

Chip Spawning: Verify all three types spawn correctly

Value Calculation: Validate formulas for each type

Fake Detection: Confirm 20% fake rate in simulator

Camera Calibration: Test color learning accuracy

Control Input: Test all keyboard commands

10.1.2 Performance Tests

Frame Rate: Maintain >30 FPS under load

Memory Leaks: Run for 30 minutes, monitor RAM

CPU Usage: Should not exceed 50% on modern hardware

10.1.3 Edge Cases

Chip Overlap: Multiple chips at same position

Screen Boundaries: Chips partially off-screen

Rapid Spawning: Burst spawn 20+ chips

Lighting Variations: Camera mode under different conditions

10.2 Validation Results

Simulator Mode:

- ■ Consistent 60 FPS
- ■ Correct value calculations (100% accuracy)
- ■ Smooth conveyor animation
- ■ All controls functional

Camera Mode:

- ■ Calibration completes in <30 seconds
- ■■ Detection accuracy depends on lighting (80-95%)
- ■ Real-time processing at 30+ FPS
- ■ Tracking maintains IDs across frames

Game Mode:

- ■ Instant chip spawning
- ■ Accurate value display
- ■ Grid provides good spatial reference
- ■ All manual controls work

11. Future Enhancements

11.1 Planned Features

11.1.1 OCR Integration

Priority: High

Effort: Medium

Implementation:

- Integrate Tesseract OCR
- Train custom model for chip digits
- Add digit validation (reject non-digits)

Benefits:

- Real digit recognition from camera
- No more random digit generation
- Production-ready authentication

11.1.2 Advanced Fake Detection

Priority: High

Effort: Medium

Methods:

Deep Learning: CNN for chip authenticity

Texture Analysis: Check for printing artifacts

Hologram Detection: Verify security features

11.1.3 Database Integration

Priority: Medium

Effort: Low

Features:

- Log all scanned chips (SQLite)
- Export to CSV/JSON
- Transaction history
- Statistics dashboard

11.1.4 Multi-Camera Support

Priority: Low

Effort: High

Concept: Multiple cameras for 360° chip scanning

11.2 Optimization Opportunities

GPU Acceleration: OpenCV CUDA support for HSV/morphology

Multi-threading: Separate capture and processing threads

Frame Skipping: Process every Nth frame for 2× speedup

ROI Processing: Only scan conveyor belt region

12. Conclusion

12.1 Project Summary

The Intergalactic Riksbanken Chip Authenticator successfully demonstrates:

Multi-mode Flexibility: Three distinct operating modes for different use cases

Real-time Performance: 30+ FPS processing with live statistics

Accurate Classification: HSV-based color detection with 80-95% accuracy

User-Friendly Interface: Intuitive controls and visual feedback

Adaptive System: Interactive calibration for varying conditions

12.2 Technical Achievements

- **■ Computer Vision Pipeline:** Complete BGR→HSV→Detection→Tracking→Display
- **■ Alpha Compositing:** Transparent PNG overlay with anti-aliasing
- **■ Value Calculation Engine:** Accurate rule-based computation
- **■ Real-time Statistics:** Live tracking of values and counts
- **■ Clean Architecture:** Modular, extensible, maintainable code

12.3 Lessons Learned

HSV vs RGB: HSV is significantly more robust for color detection

Calibration Importance: User-specific calibration improves accuracy

Alpha Blending: Proper alpha handling is crucial for visual quality

Error Handling: Graceful degradation improves user experience

Modularity: Separate modes allow independent testing and development

12.4 Applications

Developed by: Group 8 - Suneela, Sara, and Abhishek

Current Use Cases:

- Educational demonstrations
- Algorithm development
- Proof-of-concept for chip authentication
- Computer vision training

Potential Extensions:

- Currency authentication systems
- Manufacturing quality control
- Retail checkout automation
- Gaming token verification

Appendices

Appendix A: Color Space Reference

HSV Color Wheel:

0° = Red
60° = Yellow
120° = Green
180° = Cyan
240° = Blue
300° = Magenta
360° = Red (wrap)

Appendix B: Performance Benchmarks

Test Environment:

- CPU: Intel i7-10700K @ 3.8GHz
- RAM: 16GB DDR4
- GPU: Not utilized (CPU-only)
- OS: Windows 11
- Python: 3.11.9

Appendix C: Code Metrics

Metric	Value
Total Lines of Code	~1,100
Total Functions	45
Total Classes	4
Code Comments	25%
Documentation	8 files

Appendix D: References

OpenCV Documentation: <https://docs.opencv.org/>

NumPy Documentation: <https://numpy.org/doc/>

HSV Color Space: https://en.wikipedia.org/wiki/HSL_and_HSV

Alpha Compositing: https://en.wikipedia.org/wiki/Alpha_compositing

STB600 Final Project Specifications (2025)

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